LAB 8: Server Configuration – DHCP, DNS, and Web Server in Cisco Packet Tracer

OBJECTIVES:

1. To understand the working mechanism of Dynamic Host Configuration Protocol (DHCP) for automatic IP address assignment.
2. To configure a Domain Name System (DNS) server to translate domain names into IP addresses.
3. To set up and configure a Web Server (HTTP) to host a website and access it through client PCs using a web browser.

THEORY:

1. DHCP (Dynamic Host Configuration Protocol)

DHCP is a network management protocol used in TCP/IP networks.

Operation:
DHCP automatically assigns IP addresses along with subnet mask, default gateway, and DNS server information to client devices. This eliminates the need for manual configuration and prevents IP address conflicts.

Application:
Used in enterprise and home networks to efficiently manage IP address allocation and simplify network administration.

2. DNS (Domain Name System)

DNS is a hierarchical naming system used to translate human-readable domain names into numerical IP addresses.

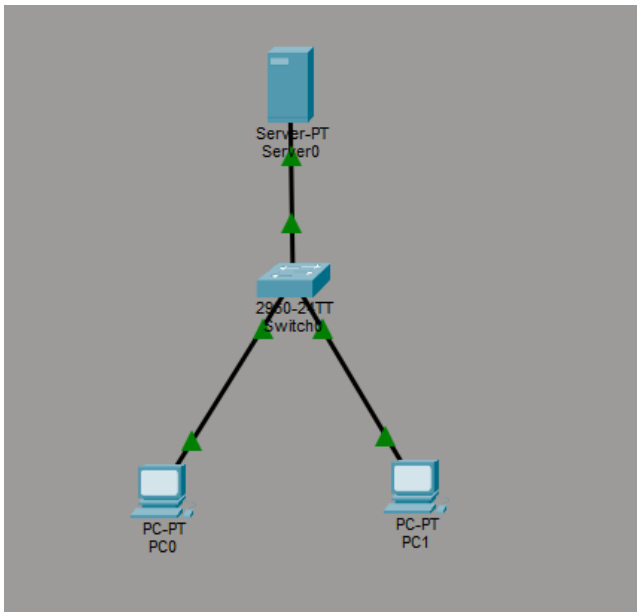
Operation:
When a user types a domain name (e.g., www.hcoe.com), the DNS server resolves it into its corresponding IP address (e.g., 192.168.1.2) so that devices can communicate.

3. Web Server (HTTP)

A Web Server stores and delivers web content to clients.

Operation:
When a client enters a URL in the browser, the server responds using HTTP protocol and sends the requested HTML page for display.

NETWORK TOPOLOGY:



- One **Server-PT (Server0)**
- One **2960-24TT Switch (Switch0)**
- Two client PCs (PC0 and PC1)

The Server acts as a centralized device providing:

- DHCP service
- DNS service
- Web (HTTP) service

All devices are connected through the switch in the same LAN network.

IP ADDRESSING SCHEME:

Device	Interface	IP Address	Subnet Mask	Default Gateway	DNS Server	Allocation Method
Server0	FastEthernet0	192.168.1.2	255.255.255.0	0.0.0.0	127.0.0.1	Static
PC0	FastEthernet0	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2	DHCP
PC1	FastEthernet0	192.168.1.4	255.255.255.0	0.0.0.0	192.168.1.2	DHCP

CONFIGURATION:

1. Server Configuration (Static IP)

The Server0 was assigned a static IP address to ensure it remains reachable by all clients.

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

This ensures the server can consistently provide DHCP, DNS, and HTTP services.

2. DHCP Configuration

The DHCP service was enabled on Server0 to dynamically assign IP addresses to PC0 and PC1.

DHCP Settings:

- Service: ON
- Pool Name: LAN_POOL
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2
- Start IP Address: 192.168.1.10
- Maximum Number of Users: 50

After enabling DHCP, client PCs were set to obtain IP addresses automatically.

3. DNS and Web Server Configuration

HTTP Configuration:

- HTTP Service: Enabled
- HTTPS Service: Enabled
- The `index.html` file was edited to display:

```
Welcome From Arpan
```

DNS Configuration:

The DNS service was enabled and a record was created:

- Name: www.hcoe.com
- Type: A Record
- Address: 192.168.1.2

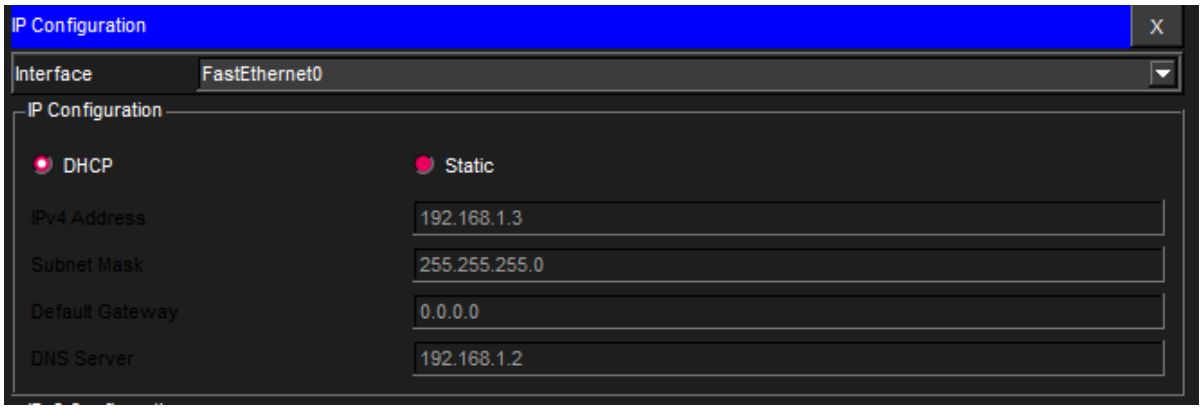
This allows the domain name to resolve to the server's IP address.

VERIFICATION:

1. DHCP Verification

PC0 and PC1 were configured to obtain IP addresses automatically using DHCP.

- IP address was successfully assigned.
- DNS server information was received automatically.
- No manual configuration was required.

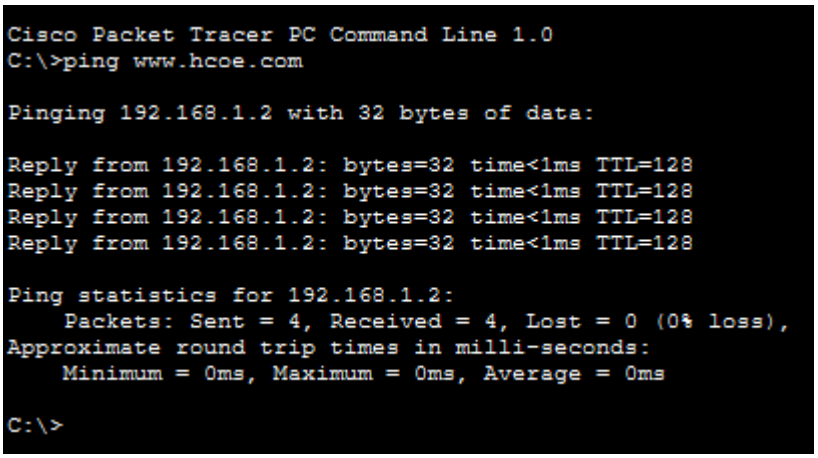


2. DNS Verification

From PC0, the following command was executed:

```
ping www.hcoe.com
```

The DNS server successfully resolved the domain name to 192.168.1.2, and replies were received without packet loss



3. Web Server Verification

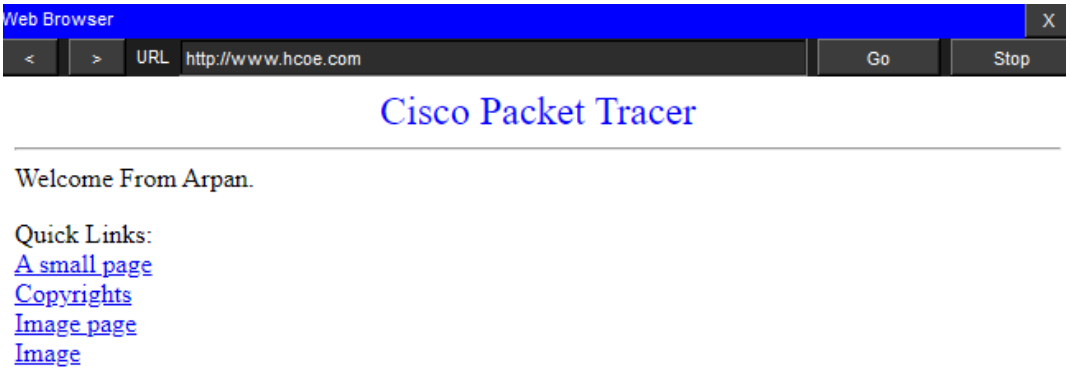
The web browser on PC0 was opened, and the URL:

`http://www.hcoe.com`

was entered.

The custom HTML page hosted on Server0 was displayed successfully, confirming that:

- DNS resolution worked correctly
- HTTP service was functioning
- The server was properly configured



RESULT:

The server was successfully configured to provide DHCP, DNS, and Web services within the LAN.

- Client PCs obtained IP addresses automatically.
- Domain name was resolved correctly.
- The hosted web page was accessed successfully through a browser.

All configurations were verified and functioned as expected.

DISCUSSION & CONCLUSION:

During this lab session, a centralized server was configured to provide DHCP, DNS, and HTTP services. We observed that the client PCs were able to obtain IP addresses dynamically from the DHCP pool, eliminating the need for manual IP configuration.

The DNS service successfully translated the domain name `www.hcoe.com` into the IP address 192.168.1.2, allowing seamless connectivity. Additionally, the Web Server delivered the custom HTML page successfully when accessed via the browser.

This lab provided practical understanding of server-based network services and their integration in a LAN environment..